

# Manufacturing Operations Data

| Operation    | Material  | Machine                           | Duty        | Shape       | Setup Time (min) | Cycle Time (sec) | Work Hours | Diameter (mm) | Length (mm) | Breadth (mm) | Height (mm) |
|--------------|-----------|-----------------------------------|-------------|-------------|------------------|------------------|------------|---------------|-------------|--------------|-------------|
| Drilling     | Steel     | Drilling machine                  | Heavy duty  | Round       | 25               | 120              | 1800       | 22            | 65          | -            | -           |
| Turning      | Aluminium | CNC                               | Medium duty | Round       | 30               | 90               | 2400       | 40            | 120         | -            | -           |
| Milling      | Steel     | 3 axis CNC                        | Heavy duty  | Rectangular | 45               | 150              | 3600       | -             | 200         | 60           | 50          |
| Grinding     | Steel     | Special Purpose Machine(SPM)      | Heavy duty  | Round       | 35               | 180              | 4000       | 25            | 80          | -            | -           |
| Jig boring   | Titanium  | CNC                               | Heavy duty  | Round       | 50               | 210              | 3200       | 18            | 90          | -            | -           |
| Rubber press | Steel     | Small (less than 650 MMX 650 MM)  | Heavy duty  | Square      | 40               | 300              | 2800       | 150           | 400         | -            | -           |
| Rubber press | Aluminium | Medium (more than 650 MMX 650 MM) | Heavy duty  | Rectangular | 55               | 280              | 3500       | 220           | 600         | -            | -           |
| Drilling     | Aluminium | Drilling machine                  | Heavy duty  | Round       | 15               | 60               | 1200       | 12            | 50          | -            | -           |
| Turning      | Steel     | CNC                               | Heavy duty  | Round       | 28               | 110              | 2600       | 55            | 180         | -            | -           |
| Milling      | Titanium  | 5 axis CNC                        | Heavy duty  | Rectangular | 60               | 240              | 4200       | -             | 300         | 70           | 65          |
| Grinding     | Steel     | Special Purpose Machine(SPM)      | Heavy duty  | Round       | 32               | 160              | 2900       | 30            | 100         | -            | -           |
| Jig boring   | Steel     | CNC                               | Heavy duty  | Round       | 48               | 200              | 3300       | 20            | 85          | -            | -           |
| Rubber press | Steel     | Medium (more than 650 MMX 650 MM) | Heavy duty  | Rectangular | 58               | 320              | 4100       | 250           | 750         | -            | -           |
| Drilling     | Steel     | Drilling machine                  | Heavy duty  | Round       | 22               | 95               | 1900       | 18            | 60          | -            | -           |
| Turning      | Aluminium | Conventional                      | Light duty  | Round       | 18               | 70               | 1500       | 35            | 140         | -            | -           |
| Milling      | Steel     | CNC                               | Medium duty | Square      | 40               | 130              | 2700       | -             | 160         | 45           | 40          |

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|--------------|-----------|-----------------------------------|------------|-------------|------------------|------------------|------------|---------------|-------------|--------------|-------------|
| Grinding     | Titanium  | Special Purpose Machine(SPM)      | Heavy duty | Round       | 38               | 190              | 3600       | 28            | 95          | -            | -           |
| Jig boring   | Steel     | CNC                               | Heavy duty | Round       | 52               | 230              | 3900       | 24            | 110         | -            | -           |
| Rubber press | Steel     | Small (less than 650 MMX 650 MM)  | Heavy duty | Square      | 42               | 260              | 3000       | 180           | 500         | -            | -           |
| Rubber press | Aluminium | Medium (more than 650 MMX 650 MM) | Heavy duty | Rectangular | 60               | 340              | 4500       | 300           | 900         | -            | -           |